

Tongtong Wu

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Research Interests

My long-term agenda is to build evolving and controllable AI systems that integrate perception, memory, reasoning and adaptation, highlighting:

- Continual learning: lifelong adaptation of language, speech, or code models with minimal forgetting.
- Knowledge graphs: dynamic graph construction and structured reasoning across modalities.
- Multi-agent systems: conditional topology design for self-evolving, resource-aware agentic collaboration.
- Code intelligence: environment-aware code generation & migration; API evolution and temporal alignment.

Research Impact

- Research impact: h-index 11, 489 citations, and 10+ A* papers in the past three years.
- Pioneered two research areas: *speech-based information extraction* and *version-controlled code generation*, with new tasks, benchmarks, metrics, and sustained methodological contributions.
- Open resources: released the *VersiCode* dataset; now part of *OpenHands*.
- Industry adoption: research supported by AUD 343K in competitive funding, has been integrated into eBay's development workflows and ByteDance's *TRAE* intelligent IDE.
- Recognition: invited keynotes and tutorials (e.g., *EMNLP*) and collaborations across North America, Europe, China, and Australia.

For details, see my 🔗 [google scholar profile](#) and 🔗 [personal website](#).

Academic Positions

Faculty of Information Technology, Monash University

Melbourne, Australia

RESEARCH FELLOW (POSTDOC)

Oct. 2022 - Present

- Working with: Prof. Reza Haffari; (past collaborator) Prof. Dragan Gasevic.

Faculty of Information Technology, Monash University

Melbourne, Australia

RESEARCH ASSISTANT ON CONTINUAL LEARNING AND LOW-RESOURCE NLP

Mar. 2021 - Jun. 2021

- Working with: Prof. Reza Haffari

Gamma AI Lab, OneConnect Smart Technology, Ping-An Group

Shanghai, China

RESEARCH ASSISTANT (INTERNSHIP) ON FEW-SHOT LEARNING

Jan. 2019 - May. 2019

- Working with: Dr. Yujin Zhu

Education

Faculty of Information Technology, Monash University

Melbourne, Australia

VISITING PH.D STUDENT, SUPERVISED BY PROF. REZA HAFFARI AND PROF. YUAN-FANG LI

Jan. 2020 - Nov. 2021

School of Computer Science and Engineering, Southeast University

Nanjing, China

PH.D. IN SOFTWARE ENGINEERING (AWARDED DEC 20, 2023), SUPERVISED BY PROF. GUILIN QI

Mar. 2018 - Dec. 2023

School of Computer Science and Engineering, Southeast University

Nanjing, China

M.S. IN SOFTWARE ENGINEERING, SUPERVISED BY PROF. GUILIN QI

Jun. 2016 - Mar. 2018

School of Internet of Things, Hohai University

Nanjing, China

B.S. IN INTERNET OF THINGS, SUPERVISED BY PROF. NING SUN

Sep. 2012 - Jun. 2016

Teaching Experience

FIT5201 - Machine Learning, Monash University

Melbourne, Australia

ROLE: GUEST LECTURER

Semester 2, 2025

- Clustering algorithms (k-means) and latent variable models (Gaussian mixture model)
- Expectation-maximization algorithms

BJSL0040 - Discrete Mathematics, Southeast University

Nanjing, China

ROLE: TEACHING ASSISTANT

Spring, 2017

- Working with Prof. Guilin Qi

Supervision & Mentoring

PhD Students

ROLE: CO-SUPERVISOR

- [2025 - Present] **Shenyu Zhang** @ Monash University (LLM-based Memory Systems)
- [2024 - Present] **Rihui Jin** @ Monash University (Tabular and Textual Reasoning)
- [2023 - Present] **Peiwen Zhao** @ Beijing Institute of Technology (Temporal Reasoning with LLM)

Master's Students

ROLE: CO-SUPERVISOR

- [2025 - Present] **Kexin Dai** @ Monash University (Agentic Recommendation Systems)
- [2025 - Present] **Nipun Nair** @ Monash University (Conversational Recommendation Systems)
- [2024 - Present] **Jiazhen Kang** @ Southeast University & Monash University (Retrieval-augmentation Code Generation)
- [2023 - Present] **Rongyi Chen** @ Southeast University & Monash University (Configurable Code Agent)

Alumni

ROLE: CO-SUPERVISOR

- [2022 - 2025] **Guitao Wang** Mphil @ Southeast University (Speech Continual Learning) -> Phd @ Monash University
- [2021 - 2024] **Jingqi Kang** Mphil @ Southeast University (Speech Information Extraction) -> ML Engineer @ iFLYTEK

Research Projects & Grants

[eBay Research Grant] - Lifelong Version-controlled Code Generation

AUD 175K

REZA HAFARI (PCI), **TONGTONG WU** (CI)

07/2025 - 07/2026

[ByteDance Research Grant] - Graph-enhanced Code Intelligence with LLMs

RMB 500K ($\approx$ AUD 106K)

GUILIN QI (PCI), **TONGTONG WU** (CI)

01/2025 - 01/2026

[Monash Seed Grant] - Towards Sustainable SE Multi-Agent Systems

AUD 30K

XIAONING DU (PCI), **TONGTONG WU** (CI)

08/2025 - 08/2026

[Monash Seed Grant] - Continual Learning of LLMs for Version-Controlled Code Generation

AUD 32K

TRANG VU (PCI), **TONGTONG WU** (CI)

08/2024 - 08/2025

[DARPA CCU] - Computational Cultural Understanding

AUD 4M

RESEARCH FELLOW (FULL TIME), WORKING WITH PROF. REZA HAFARI

10/2022 - 07/2024

[DARPA KMASS] - Large-scale Multi-modal Knowledge Management

AUD 3M

RESEARCH FELLOW (PART TIME, 0.5 FTE), WORKING WITH PROF. DRAGAN GASEVIC

10/2022 - 02/2023

Selected Publications

K-DeCore: Facilitating Knowledge Transfer in Continual Structured Knowledge Reasoning via Knowledge Decoupling

NeurIPS

YONGRUI CHEN, YI HUANG, Y. LIU, SHENYU ZHANG, JUNHAO HE, **TONGTONG WU**, GUILIN QI, TIANXING WU

2025

- Core Rank - A*

Can LLMs Evaluate Complex Attribution in QA? Automatic Benchmarking Using Knowledge Graphs

ACL

NAN HU, JIAOYAN CHEN, Y. WU, GUILIN QI, H. WANG, SHENG BI, YONGRUI CHEN, **TONGTONG WU**, JEFF Z PAN

2025

- Core Rank - A*

Continual Speech Learning with Fused Speech Features

Interspeech

GUITAO WANG, JINMING ZHAO, HAO YANG, GUILIN QI, **TONGTONG WU***, GHOLAMREZA HAFFARI

2025

- Core Rank - A, * Corresponding Author

Continual Multimodal Knowledge Graph Construction

IJCAI

XIANG CHEN, JINTIAN ZHANG, XIAOHAN WANG, NINGYU ZHANG, **TONGTONG WU**, CHENGMING WANG, FEIYU XIONG,

2024

WEI GUO AND HUAJUN CHEN

- Core Rank - A*

Towards LifeSpan Cognitive Systems

Trans. Mach. Learn. Res.

YU WANG*, CHI HAN*, **TONGTONG WU***, XIAOXIN HE*, WANGCHUNSHU ZHOU, NAFIS SADEQ, XIUSI CHEN, ZEXUE

2024

HE, WEI WANG, GHOLAMREZA HAFFARI, HENG JI, JULIAN MCAULEY

- * Equal Contribution

KCG-EE: Effective Knowledge Conditioning for Generation-based Event Extraction

World Wide Web J.

TONGTONG WU, FATEMEH SHIRI, JINGQI KANG, GUILIN QI, YUAN-FANG LI, REZA HAFFARI

2023

- Core Rank - A

Pretrained Language Model in Continual Learning: A Comparative Study

ICLR

TONGTONG WU, MASSIMO CACCIA, ZHUANG LI, YUAN-FANG LI, GUILIN QI, REZA HAFFARI

2022

- Core Rank - A*

Towards Relation Extraction From Speech

EMNLP

TONGTONG WU*, GUITAO WANG*, JINMING ZHAO*, Z. LIU, GUILIN QI, YUAN-FANG LI, GHOLAMREZA HAFFARI

2022

- Core Rank - A*, * Equal Contribution

Curriculum-Meta Learning for Order-robust Continual Relation Extraction

AAAI

TONGTONG WU, XUEKAI LI, YUAN-FANG LI, REZA HAFFARI, GUILIN QI, YUJIN ZHU, GUOQIANG XU

2021

- Core Rank - A*

Selected Under Review

VersiCode: Towards Version-controllable Code Generation

TONGTONG WU, WEIGANG WU, XINGYU WANG, KANG XU, SUYU MA, BO JIANG, PING YANG, ZHENCHANG XING,

YUAN-FANG LI, GHOLAMREZA HAFFARI

- New Benchmark

Environment-Aware Code Generation: How far are We?

TONGTONG WU, RONGYI CHEN, WENJIE DU, SUYU MA, GUILIN QI, ZHENCHANG XING, SHAHRAM KHADIVI, RAMESH

PERIYATHAMBI, GHOLAMREZA HAFFARI

- Empirical Study, Conditional Accepted by ICSE 2026, Core Rank - A*

GraphCode: Structuring API Evolution for LLM-based Code Migration

TONGTONG WU, JIAZHEN KANG, YUCHEN LU, GUILIN QI, SUYU MA, KANG XU, NINGYUAN SUN, BO JIANG,
ZHENCHANG XING, GHOLAMREZA HAFFARI

- New Method

CARD: Towards Conditional Design of Multi-agent Topological Structures

TONGTONG WU, YANMING LI, ZIYE TANG, CHEN JIANG, LINHAO LUO, GUILIN QI, SHIRUI PAN, GHOLAMREZA HAFFARI

- New Method

Selected Patents

A Document-level Event Extraction Method Based On Task-conditioned Generation

Chinese Patent

TONGTONG WU, JINGQI KANG, GUILIN QI

2022

- Granted (202210403218.6)

A Continual Relation Extraction Method Based on Task-aware Meta Learning

Chinese Patent

TONGTONG WU, XUEKAI LI, GUILIN QI

2021

- Granted (202110099305.2)

Professional Service

Member of Program Committee

SELECTED CONFERENCES

- ICML [2023 - 2025], NeurIPS [2023 - 2025], ICLR [2023 - 2026]
- ACL ARR [2022 - 2025], AAI [2022, 2024 - 2026], IJCAI [2023 - 2024]

Reviewer of Journals

SELECTED JOURNALS

- TKDE (IEEE Trans. Knowl. Data Eng.), TCSS (IEEE Trans. Comput. Soc. Syst.)
- TCSVT (IEEE Trans. Circuits Syst. Video Technol.), TIP (IEEE Trans. Image Process.)
- TIST (ACM Trans. Intell. Syst. Technol.), TALLIP (ACM Trans. Asian Low Resour. Lang. Inf. Process.)
- KBS (Knowl. Based Syst.)

Editorial Board Member of Journals

SELECTED JOURNALS

- DI (Data Intelligence) [IF - 2.31]

Awards & Scholarships

Jun. 2020 **Chinese Government**, Chinese Scholarship Council Scholarship

Beijing, China

Sep. 2018 **Southeast University**, Outstanding Postgraduates

Nanjing, China

Jun. 2016 **Hohai University**, Hohai Excellent Graduates - Top 50

Nanjing, China

Sep. 2014 **Hohai University**, Outstanding Student Pacesetter

Nanjing, China

Sep. 2013 **Chinese Government**, National Encouragement Scholarship

Nanjing, China

Community Engagement

Nov 2025 **Continual Learning of Large Language Models**, (3-hr) Tutorial Speaker & Organizer @ EMNLP

Suzhou, China

Nov 2024 **Continual Learning of Large Language Models**, (3-hr) Tutorial Speaker & Organizer @ AJCAI

Melb., Australia

Oct 2024 **The Advances of Continual Learning with LLMs**, Keynote Speaker of CL workshop @ ACM-MM

Melb., Australia

Aug 2024 **Towards Version-Controllable Code Generation**, Invited Speaker @ Oracle AI Research Seminar

Melb., Australia

Mar 2022 **Continual Learning with Pretrained Language Models**, Invited Speaker @ AI-TIME

LIVE Online